Homework 19: Invertible Linear Transformations

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BCS A, E, F, G

Problem 1. An $m \times n$ upper triangular matrix is one whose entries below the main diagonal are 0's. When is a square upper triangular matrix invertible? Justify your answer.

Problem 2. An $m \times n$ lower triangular matrix is one whose entries above the main diagonal are 0's. When is a square lower triangular matrix invertible? Justify your answer.

Problem 3. If A is an $n \times n$ matrix and the transformation $\boldsymbol{x} \to A\boldsymbol{x}$ is one-to-one, what else can you say about this transformation? Justify your answer.

Problem 4. Consider the linear transformation $T: \mathbb{R}^n \to \mathbb{R}^m$ represented by $T(\boldsymbol{x}) = A\boldsymbol{x}$. Find the nullity and rank of T, and determine whether T is one-to-one, onto, or neither.

a. $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$.

b. $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$.

c. $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$.

d. $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \end{bmatrix}$.

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Good Luck